Diabetes Writeup

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The goal of this study is to see if there is a difference in the treatment effects of three different approaches to prevent pre diabetic patients from progressing into obtaining full diabetes. The three different approaches were Metformin (Met), Rosiglitazone (Rosi), and Lifestyle. Lifestyle approach includes focusing on a good diet and exercise. Since we are doing a three sample comparison with limited data, the analysis uses a bootstrapped sample to create a kernel density estimator for the distribution of each approach and then we use a Tukey's post hoc approach to see if there are any differences in the CI's of the sample by applying a Bonferroni-Adjustment. The sample has 699 patients. 134 patients were given the Met treatment, 133 patients were given the Rosi treatment and 132 patients were given the Lifestyle treatment. The total amount of successes (patient did not become diabetic) and failures (patient became diabetic) were recorded for each patient. These values can be seen in Table 1 below.

outcome	failure	success
treatment		
lifestyle	109	125
met	120	112
rosi	90	143

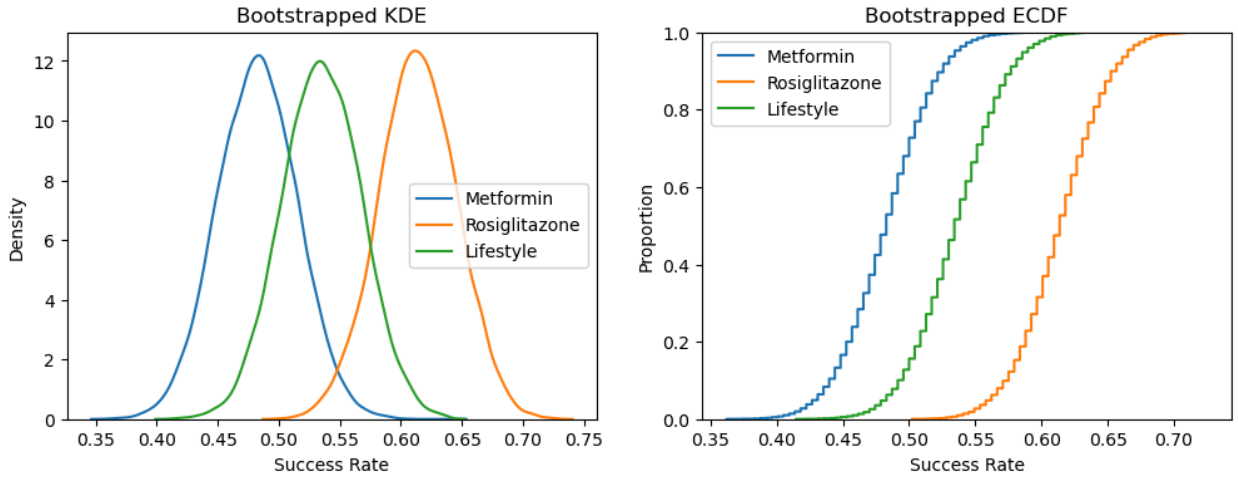
treatment	
lifestyle	0.534
met	0.483
rosi	0.614

Name: success, dtype: float64

Table 1

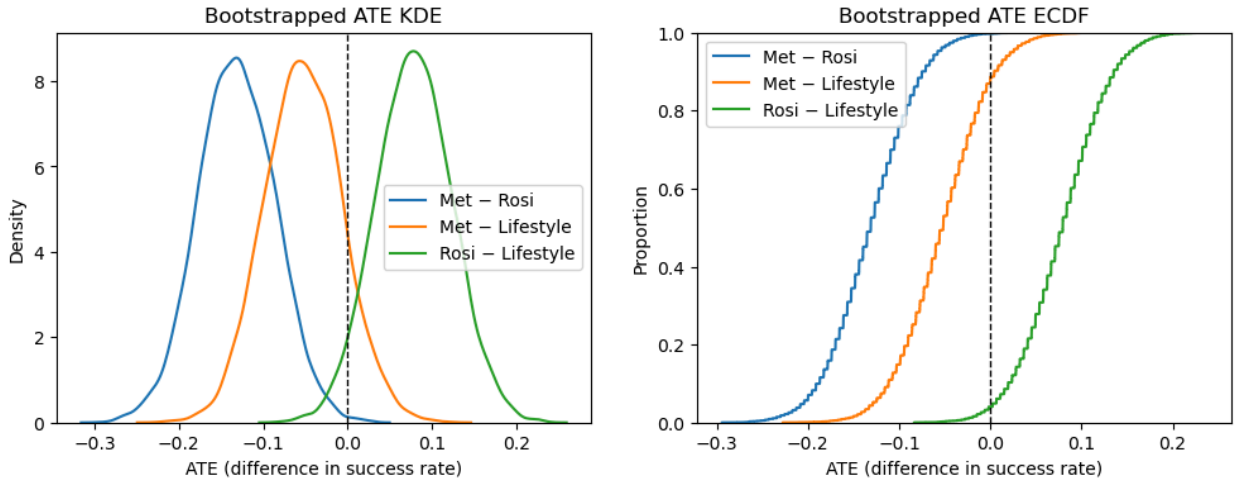
Table 1 shows that from the sample Rosi was the most successful with just over 61 percent of patients not progressing to full diabetes and met was the least successful approach with just over half of the patients progressing to full diabetes. To learn more information about the behavior of this data and see which of these approaches are statistically different from each other, bootstrapping techniques were used to create a kernel density estimator distribution for each approach. The KDE measured the success rate of the specified approach. Empirical cumulative distribution functions

were created to provide more context about the step by step cumulative changes to the distribution. Graph 1 shows the KDE on the left and the ECDF on the right.



Graph 1

Both the KDE and the ECDF confirm what we learned from table 1 showing that Rosi is the most successful approach from the sample and Met is the least successful approach. Since all three approaches are bootstrapped from the same sample the KDE and the ECDF's are symmetrical. To see if the difference between each sample is significant we need to create KDE's and ECDF's for the three Tukey's two-sample comparisons. Graph 2 shows the KDE for the comparisons on the left and ECDF on the right. Instead of the success rate on the x-axis, the Average Treatment Effect (difference between the success rate of specified comparisons) is displayed. A horizontal dotted line along a treatment effect of 0 is used to make the graphs more comprehensive.



Graph 2

Graph 2 has the very similar shapes to graph 1 for both the KDE and the ECDF's. This is because each approach has identical distribution shapes. When we take KDE of Met - KDE ROSI since the shape of KDE Met and KDE Rosi are the same, the shape of KDI (MET - ROSI) will be the same. Rosi - Lifestyle has a positive ATE because the sample success rate for Roti is larger than the sample proportion for Lifestyle. Since Met has the lowest sample proportion, MET - ROSI and MET - Lifestyle have negative ATE's. To find if the differences are statistically significant, we need to focus on the CI's of our ATE. The ECDF'S can give us more context about the nature of the CI's. Looking at where the ECDF's intersects 0 for each distribution can show us how likely a difference is to be significant. Met - Rosi intersects 0 at approximately 1 meaning that the chances of the true population success rate of Met being the same or larger than Rosi is 1 - approximately 1 or just approximately 0. Met - Lifestyle appears to intersect 0 near 0.90. This shows us that the percent chance that the population success rate for Met being the same or larger than Lifestyle is approximately 0.1. The last one is Rosi - Lifestyle. Since Rosi's sample proportion is larger than Lifestyle's we see the ECDF intersect 0 much earlier in the CDF at approx 0.05. So the probability that Rosi has a population success rate that is the same or less than Lifestyle is approximately 0.05. These numbers are just estimates from the graph, but we can pull the specific CI's at different confidence intervals. Table 2 shows the confidence intervals for the three comparisons.

Comparison	ATE	90% CI	95% CI	99% CI
Met – Rosi	-0.131	[-0.204, -0.058]	[-0.221, -0.041]	[-0.251, -0.015]
Met – Lifestyle	-0.053	[-0.129, 0.026]	[-0.142, 0.039]	[-0.172, 0.069]
Rosi – Lifestyle	0.079	[0.007, 0.152]	[-0.010, 0.169]	[-0.036, 0.195]

Table 2

When we run a Tukey's comparison test we have a multiple testing problem. Using an alpha of 0.05 (95% CI) is the standard in statistics. However, when we have multiple tests we have to make an adjustment to our alpha to reduce the likelihood of a type 1 error. If we have one test the theoretical limit for a type 1 error is held to 0.05, but if we have three tests the odds of a type 1 error increases. A Bonferroni adjustment makes the alpha stricter for each test. The alpha star (multiple tests alpha) is our original alpha divided by the number of tests. This keeps the summation of each alpha at our original alpha (0.05). This makes our alpha star 0.05/3 or 0.016. To adjust this to our CI we just take 1 - alpha star (0.984). For simplicity, we are just going to use the 99% confidence interval. In this interval we see that 0 is within the interval for Met - Lifestyle and Rosi - Lifestyle. This tells us that the

difference in population success rate for Met and Rossi approaches is not significantly different than the population success rate for changes in Lifestyle as a method to prevent diabetes. We fail to reject the null hypothesis for these two comparisons. The confidence interval for Met and Rossi does not contain 0, so the population success rate for Rossi is significantly larger than the population success rate for the Met approach at preventing diabetes. We can reject the null hypothesis that the success rates are not different at an alpha of 0.05 and an alpha star of 0.016.